

Yuchen Li

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Summary

I am the first author of 3D-CoMPaT, which was presented as an oral paper at ECCV, and an extended version, 3D-CoMPaT++, was accepted at TPAMI 2025. I also led the NeurIPS paper PointNeXt, which has received 1,350+ citations. I have served as a core organizer of a CVPR Workshop and as a program chair and reviewer for leading AI conferences and journals such as TPAMI, IJCV, CVPR, ICCV, AAAI, TCSVT and NeurIPS. I conducted a research internship at Amazon Science, Seattle, and Sony AI, Tokyo. My research focuses on Multimodal Visual Language Models (VLM) for image, audio, video, 3D, and 4D generation and perception.

Education

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI) <i>PhD in Computer Vision</i>	Aug 2023 – May 2027 <i>Abu Dhabi, UAE</i>
King Abdullah University of Science and Technology (KAUST) <i>M.Sc. in Computer Science (Visual Computing Center)</i> GPA: 3.67/4.00, Thesis: Compositional and Low-shot Understanding of 3D Objects	Aug 2020 – May 2022 <i>Thuwal, Saudi Arabia</i>
Southern University of Science and Technology (SUSTech) <i>B.Sc. in Computer Science and Technology</i> GPA: 3.6/4.0, Thesis: Few-Shot Learning on 3D Object Recognition	Aug 2017 – Jun 2021 <i>Shenzhen, China</i>
National University of Singapore (NUS) <i>School of Computing (Exchange Student)</i>	Jan 2020 – May 2020 <i>Singapore</i>
University of British Columbia (UBC) <i>Electronics and Computer Engineering (Visiting Student)</i>	Jul 2018 – Aug 2018 <i>Vancouver, Canada</i>

Research Experience

Amazon Science - Prime Video <i>Applied Scientist Research Intern</i> Scaled distributed training for Perceptio, a perception-augmented VLM, utilizing 256 NVIDIA H200 GPUs on AWS to enable complex 2D and 3D spatial perception reasoning	July 2025 – Nov 2025 <i>Seattle, USA</i>
Sony AI Research <i>Research Intern</i> Developed music-driven dance video generation pipeline, curating high-quality motion datasets to train a diffusion-based video model for music-driven dance generation	Jun 2024 – Nov 2024 <i>Tokyo, Japan</i>
Mohamed bin Zayed University of Artificial Intelligence <i>Assistant Researcher</i> Explored 3D and Video content generation and perception using diffusion and LLM models	Sep 2023 – Present <i>Abu Dhabi, UAE</i>
KAUST - Vision CAIR Group <i>Research Assistant</i> Researching semi-supervised few-shot learning, meta-learning, and 3D object recognition	Aug 2020 – Aug 2022 <i>Thuwal, Saudi Arabia</i>
Tsinghua University - Digital Manufacturing Lab <i>Visiting Research Student</i> Led a team to research Micro-Organic Prediction Modeling of Cladding Process using CNNs	Jun 2019 – Jul 2019 <i>Beijing, China</i>

Professional Experience

Dubai Business Associates - Dubai Future Foundation / Emirates Airlines Sep 2022 – Jun 2023
Innovation Associate *Dubai, UAE*

- Selected for a prestigious mini-MBA (0.3% acceptance); consulted for Emirates Airlines to deliver data-driven AI chatbot and 3-year strategic performance roadmap presented to VPs

Alibaba - Summer of Code Student Developer @ Apache Mar 2019 – Jul 2019
Student Developer *Remote*

- Developed the Apache RocketMQ JDBC Connector (23 commits, 3.4k lines of code) as an independent developer, contributing an external project to Apache RocketMQ (22k+ stars)

Publications & Spotlight Talks

1. **Rethinking Music-Driven Dance Generation in 2D: Long-Horizon Pose Diffusion for Video Synthesis.** Yuchen Li (First Author) et al. Under Review.

2. **Perceptio: Perception Enhanced Vision Language Models via Spatial Token Generation.** Yuchen Li, Amanmeet Garg, Shalini Chaudhuri, Rui Zhao, Garin Kessler. Arxiv 2603.18795

3. **3DCoMPaT++: An improved Large-scale 3D Vision Dataset for Compositional Recognition.** Habib Slim, Xiang Li, Yuchen Li, Mahmoud Ahmed, Mohamed Ayman, Ujjwal Upadhyay, Ahmed Abdelreheem, Arpit Prajapati, Suhail Pothigara, Peter Wonka, Mohamed Elhoseiny, IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2025.

4. **Exploring Scaling Laws of PointNets.** Yuchen Li*, Guocheng Qian*, Houwen Peng, Jinjie Mai, Hasan Abed Al Kader Hammoud, Mohamed Elhoseiny, Bernard Ghanem, International Conference on 3D Vision (3DV, Nectar Track Spotlight Talk), 2025.

5. **PointNeXt: Revisiting PointNets with Improved Training and Scaling Strategies.** Yuchen Li*, Guocheng Qian*, Houwen Peng, Jinjie Mai, Hasan Abed Al Kader Hammoud, Mohamed Elhoseiny, Bernard Ghanem, Conference on Neural Information Processing Systems (NeurIPS), 2022.

6. **3D CoMPaT Dataset: Composition of Materials on Parts of 3D Things.** Yuchen Li*, Ujjwal Upadhyay*, Habib Slim, Tezuesh Varshney, Ahmed Abdelreheem, Arpit Prajapati, Suhail Pothigara, Peter Wonka, Mohamed Elhoseiny, European Conference on Computer Vision (ECCV, Oral Presentation), 2022.

7. **Semi-Supervised Few-Shot Learning with Prototypical Random Walks.** Ahmed Ayyad, Yuchen Li, Raden Muaz, Shadi Albarqouni*, Mohamed Elhoseiny*, AAAI Meta Learning workshop (Oral Presentation), 2021.

Honors & Awards

Organizer of Workshop on Compositional 3D Vision & 3DCoMPaT Challenge Jun 2023
CVPR 2023 *Vancouver, Canada*

Super AI Youth from Jiangmen Innovation Ventures May 2021
Jiangmen Innovation Ventures *Beijing, China*

Honor Mention, International Interdisciplinary Contest in Modeling (Top 16%) Apr 2019
International Interdisciplinary Contest in Modeling *Shenzhen, China*

2nd Prize of ASC World University Student Supercomputer Challenge (Top 10%) Jun 2019
ASC World University Student Supercomputer Challenge *Shenzhen, China*